Title: Site-to-Site IPsec VPN — Port Filtering Practical

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Objective

"The objective of this practical is to configure a site-to-site IPsec VPN tunnel between two OPNsense firewalls and apply port-based firewall rules to allow only specific traffic (HTTP, HTTPS, SSH, RDP, FTP, DNS) through the tunnel while blocking all other traffic."

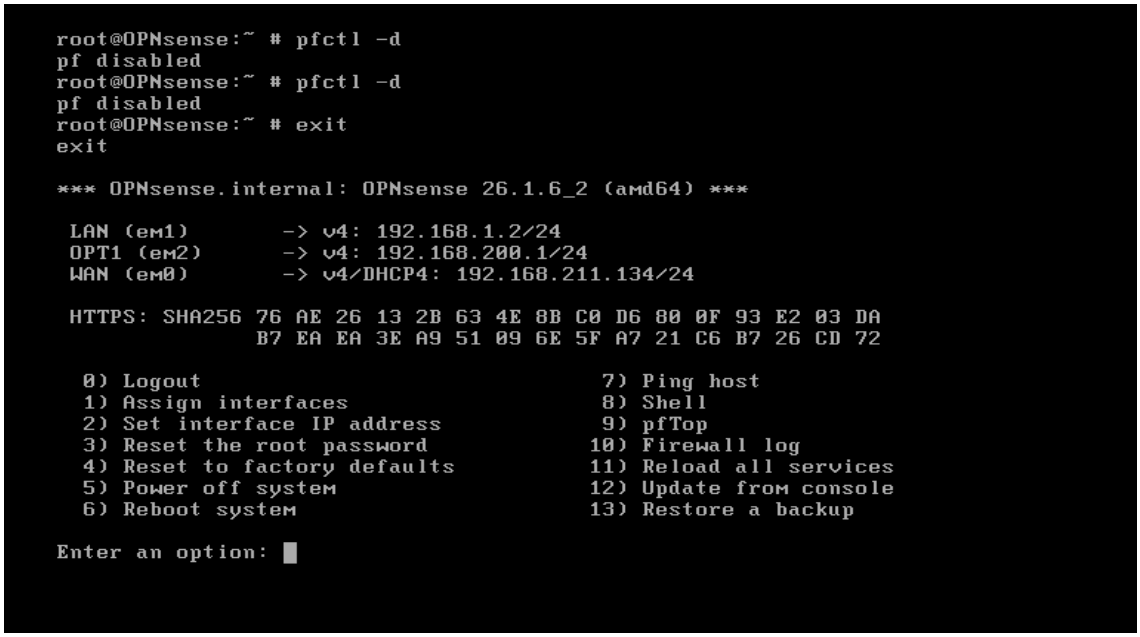
Lab Topology & IP Plan

Device	Interface	IP Address
OPNsense A	WAN	192.168.211.133
OPNsense A	LAN	192.168.1.1
OPNsense A	LAB_NET	192.168.100.1
OPNsense B	WAN	192.168.211.134
OPNsense B	LAN	192.168.2.1
OPNsense B	LAB_NET	192.168.200.1
Kali VM	LAB_NET	192.168.100.100

Site: A



Site: B

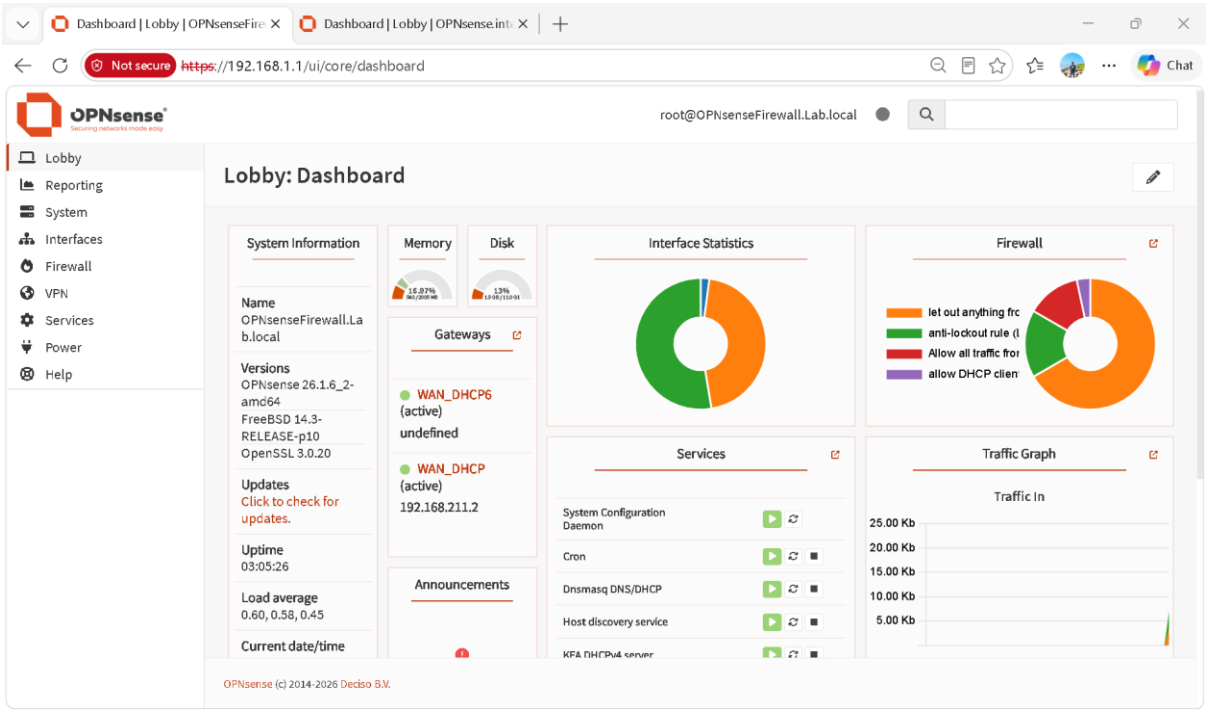


Section 4 — Understanding the Ports

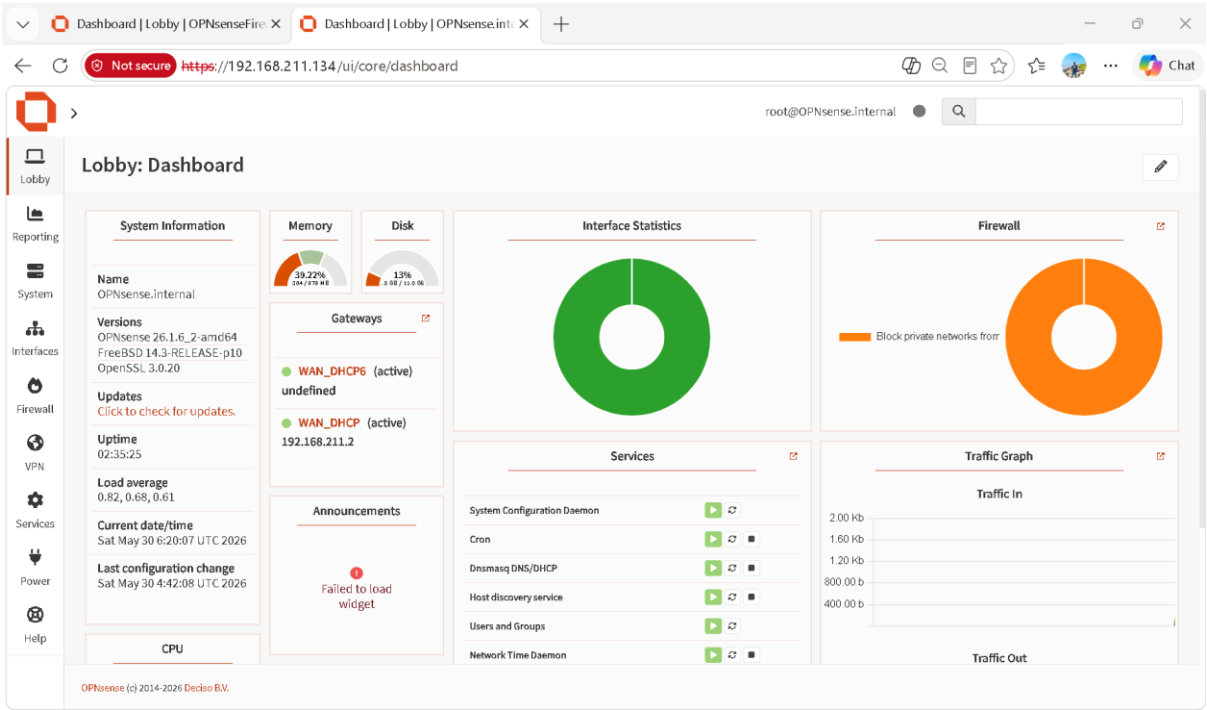
Port	Protocol	Service	Used For
80	TCP	HTTP	Normal websites
443	TCP	HTTPS	Secure websites
22	TCP	SSH	Remote terminal access
3389	TCP	RDP	Windows Remote Desktop
21	TCP	FTP	File transfer
53	TCP/UDP	DNS	Domain name resolution

Both Firewall Gui Dashboard:

Site A -



Site -B.



Section 5 — Firewall Rules on Site A

OPEN : **https://192.168.1.1**

Then - **Firewall → Rules → IPsec**

Note : This is where we control traffic **coming through the tunnel**.

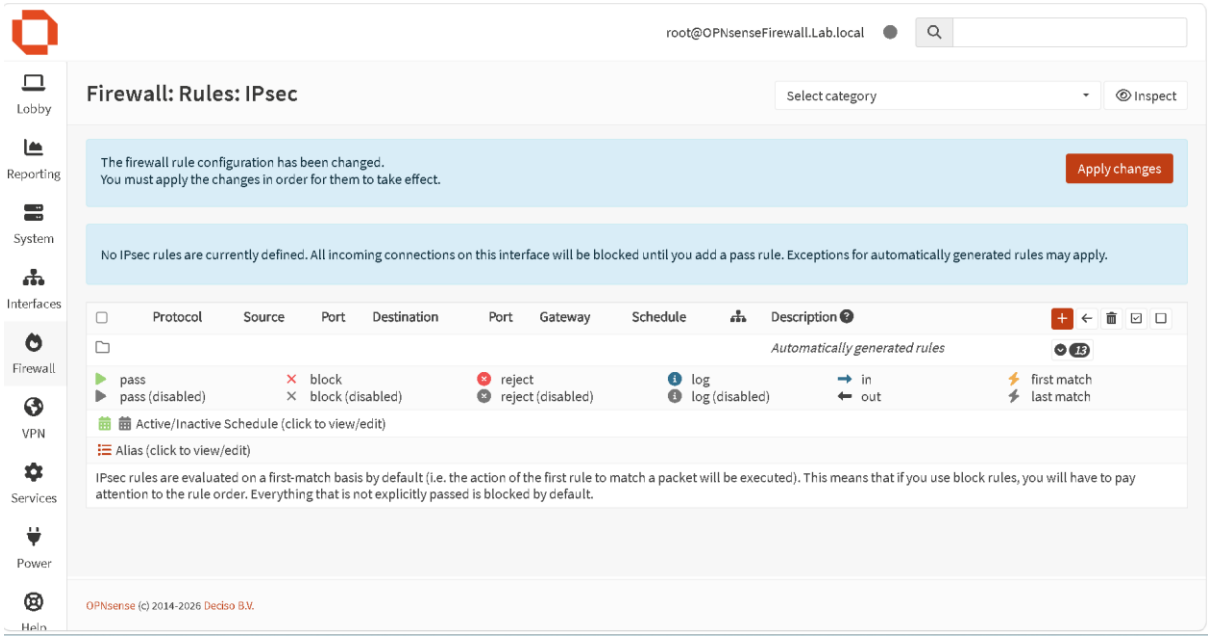


Fig . Currently no rule exists , now I am going to create rule one by one

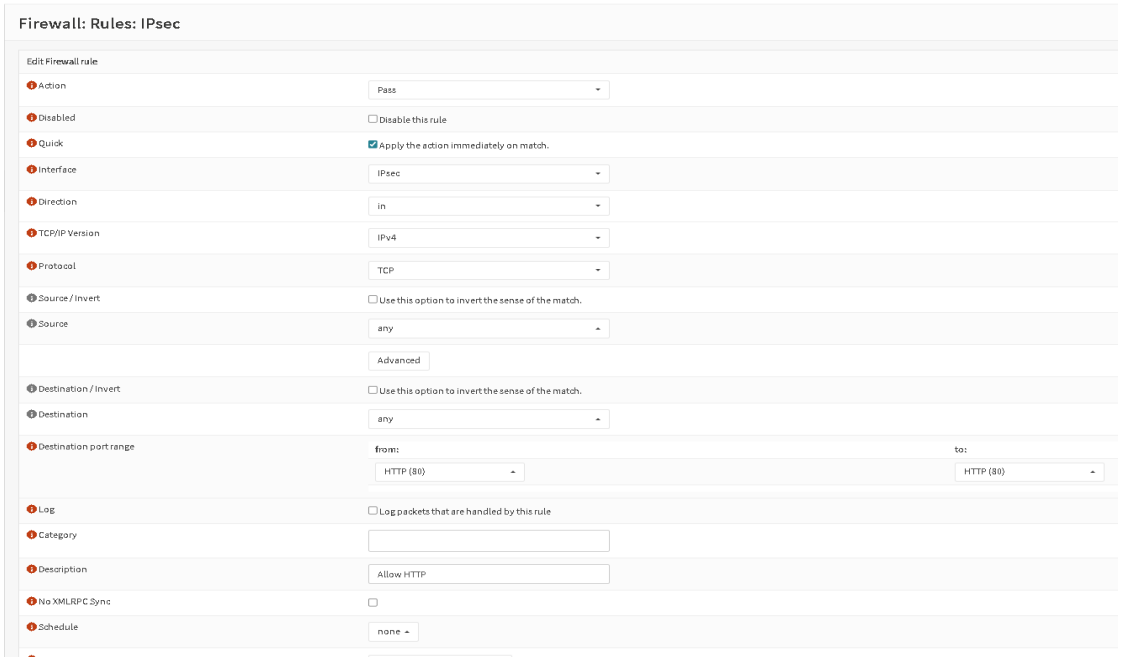


Fig . the above I creating rule to allow port 80

Rule Created:

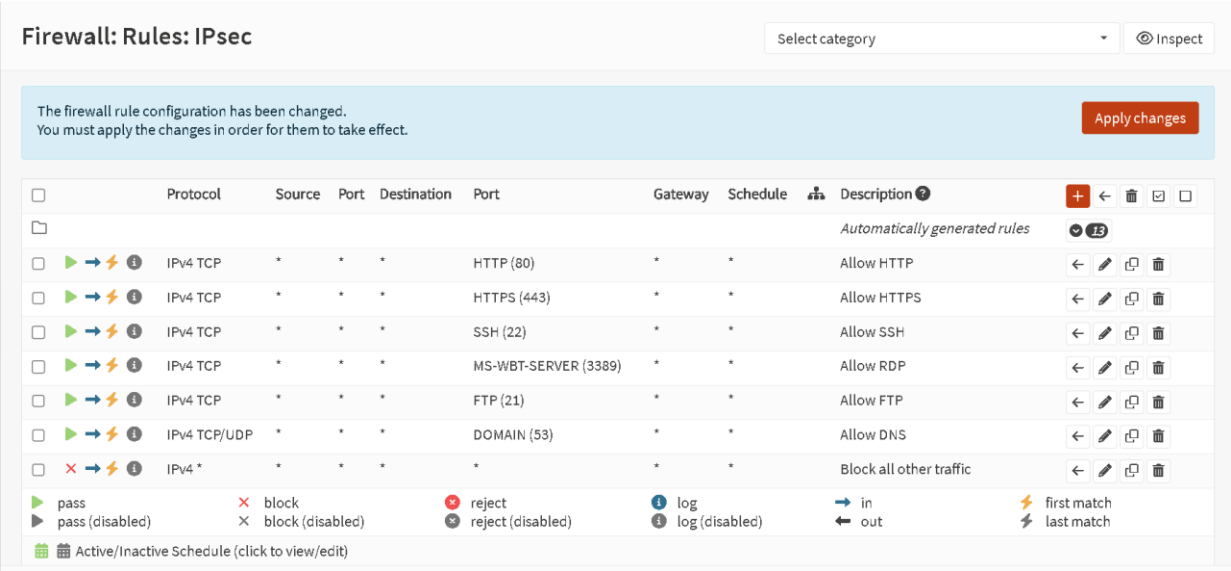


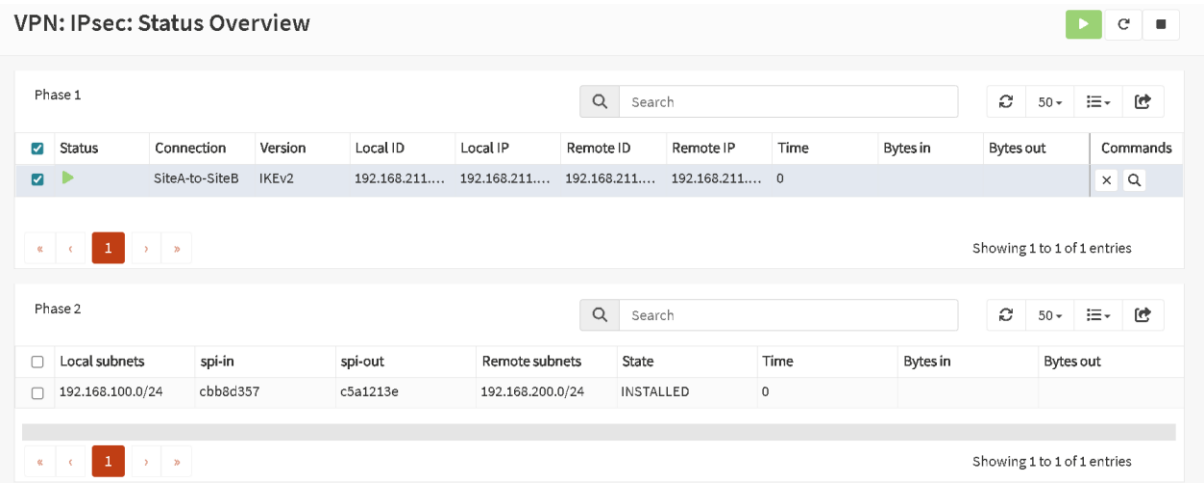
Fig. the last rule is created to block all port instead above

"Note:

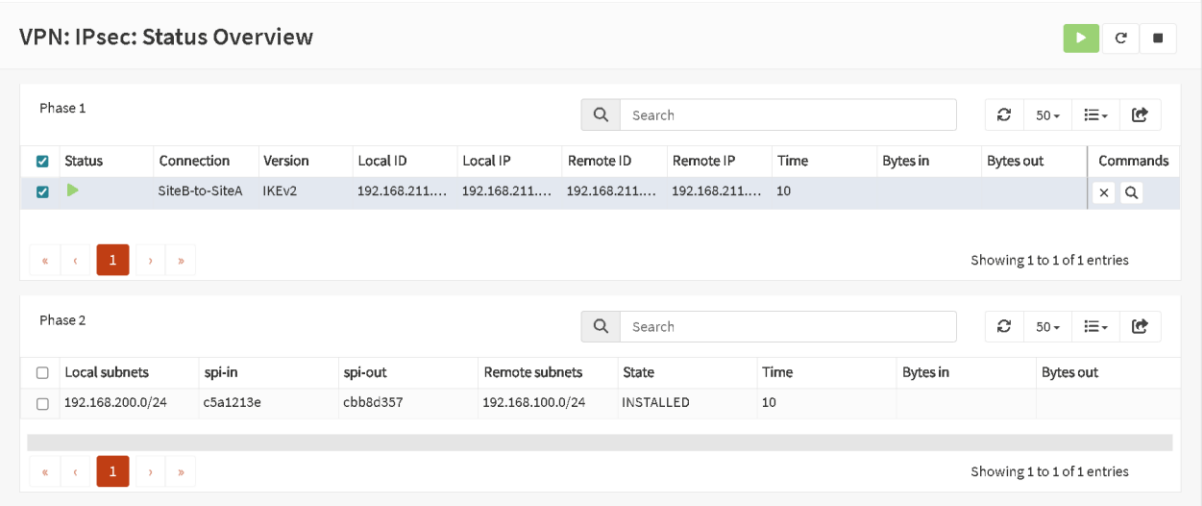
OPNsense B GUI is accessed via WAN IP (192.168.211.134) due to lab limitations in VMware Player free version. In a production environment, firewall management is always performed through a dedicated management interface or LAN interface. The pfctl -d command was used temporarily to allow GUI access during configuration. All firewall rules and IPsec configuration are saved permanently in the configuration file."

IPsec Status Overview (tunnel established):

SITE A:



SITE B:



Ping to check the Ipsec tunnel is working and target is alive :

```
┌──(kali㉿kali)-[~]
└─$ ping 192.168.200.1
PING 192.168.200.1 (192.168.200.1) 56(84) bytes of data.
64 bytes from 192.168.200.1: icmp_seq=1 ttl=63 time=5.11 ms
64 bytes from 192.168.200.1: icmp_seq=2 ttl=63 time=1.11 ms
64 bytes from 192.168.200.1: icmp_seq=3 ttl=63 time=3.52 ms
64 bytes from 192.168.200.1: icmp_seq=4 ttl=63 time=1.75 ms
^C
── 192.168.200.1 ping statistics ──
4 packets transmitted, 4 received, 0% packet loss, time 3000ms
rtt min/avg/max/mdev = 1.106/2.872/5.114/1.567 ms
```

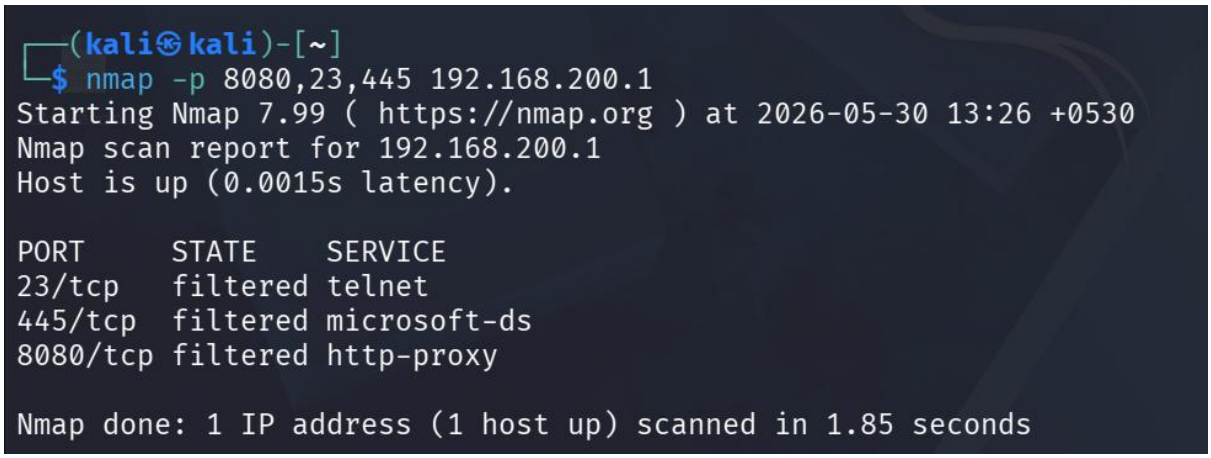
Nmap results (allowed ports):

```
┌──(kali㉿kali)-[~]
└─$ nmap -p 80,443,22,3389,21,53 192.168.200.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-30 13:24 +0530
Nmap scan report for 192.168.200.1
Host is up (0.0030s latency).

PORT     STATE    SERVICE
21/tcp   filtered ftp
22/tcp   filtered ssh
53/tcp   open     domain
80/tcp   open     http
443/tcp  open     https
3389/tcp filtered ms-wbt-server

Nmap done: 1 IP address (1 host up) scanned in 1.86 seconds
```

Nmap results (blocked ports):



Conclusion

"This practical successfully demonstrated the configuration of a site-to-site IPsec VPN tunnel between two OPNsense firewalls using IKEv2 protocol with Pre-Shared Key authentication. Port-based firewall rules were applied on the IPsec interface to allow only specific services (HTTP, HTTPS, SSH, FTP, DNS, RDP) while blocking all other traffic. Testing using Nmap confirmed that allowed ports are accessible through the tunnel and all other ports are blocked as expected. This configuration simulates a real-world enterprise network where only necessary services are permitted over VPN tunnels for security purposes."

PRACTICAL: 2

Title: Hub and Spoke IPsec VPN — Theoretical Design

Topology:

```
Site A ⟵ IPsec Tunnel 1 ⟶ Site B ⟵ IPsec Tunnel 2 ⟶ Site C
(India)                    (Singapore)                  (Japan)
```

IP Plan:

Device	WAN	LAN	LAB_NET
OPNsense A	192.168.211.133	192.168.1.1	192.168.100.1
OPNsense B	192.168.211.134	192.168.2.1	192.168.200.1
OPNsense C	192.168.211.x	192.168.3.1	192.168.300.1

Tunnel Configuration:

Tunnel	Local	Remote	Child SA Local	Child SA Remote
A↔B	192.168.211.133	192.168.211.134	192.168.100.0/24	192.168.200.0/24
B↔C	192.168.211.134	192.168.211.x	192.168.200.0/24	192.168.300.0/24

Why A cannot talk to C:

There is no IPsec tunnel configured between Site A and Site C. Site A's routing table has no entry for 192.168.300.0/24 network. Any packet from Site A destined for Site C will be dropped at OPNsense A because no matching Child SA exists.

Lab Limitation:

Site C was not practically implemented due to insufficient RAM in the lab environment. The configuration would follow identical steps as the A↔B tunnel with different IP ranges.